6th International Innovation Competition (ITC-EGYPT 2026) Graduation Projects' Template

Faculty/school: Faculty of Computer Science and Artificial Intelligence
University / Studying administration: Damietta University

Nabd: An AI-Driven Ecosystem for Maternal Health Monitoring and Risk Detection

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Executive Summary

Nabd is an AI-driven healthcare ecosystem designed to mitigate pregnancy-related risks and enhance maternal safety. The platform integrates a high-performance .NET (ASP.NET Core MVC) back-end with Python-based FastAPI microservices to deliver real-time diagnostic assistance. The system features three primary AI components: an automated OCR pipeline for lab reports and ingredient analysis, a Gestational Diabetes Mellitus (GDM) prediction engine, and a multi-class Maternal Health Risk Detection model. Beyond automated analysis, Nabd provides a comprehensive monitoring framework that facilitates direct communication between doctors and patients, ensuring continuous clinical supervision.

Pregnancy is a critical life stage, and focus on maternal and fetal health has grown significantly. Despite this, health risks are rising; a 2021 study [1] found that 14% of pregnant women suffer from gestational diabetes (GDM), which is directly linked to diabetes and cardiovascular diseases. Furthermore, the World Health Organization (WHO) reported in 2023 [2] an alarming increase in maternal mortality and health risks. Beyond medical conditions, concerns regarding skincare and body products are significant. Organizations like the American College of Obstetricians & Gynecologists (ACOG) [3] have warned against ingredients such as retinoids. Yet, many products lack clear safety information for expectant mothers, and the communication gaps include a lack of integrated platforms that provide seamless, real-time communication between doctors and patients for proactive decision-making.

The project is highly feasible for practical implementation, as the core architecture and models are already developed:

- Proven Accuracy: The maternal risk detection model achieved 98% accuracy, and the GDM engine achieved 93.52% accuracy.
- Scalable Architecture: Built on a modular three-tier framework (Presentation, Business Logic, and Data Access Layers) using ASP.NET Core and MVC.
- Functional Prototypes: The system already features functional dashboards for Patients, Doctors, Assistants, and Admins.

Currently, the project has been developed through independent research and technical execution; there are no external financiers or corporate partners involved at this stage.

The project adds many important features to this field of industry:

- Active Prediction: Unlike passive logging tools like Flo [4], Nabd employs an active prediction layer to identify risks in pregnancy or disease like GDM before they escalate.
- Dynamic OCR Analysis: While tools like Yuka [5] rely on fixed barcode databases, Nabd uses flexible OCR (docTR) to interpret any medical report or ingredient label dynamically.
- Clinical Decision Support: It acts as a bridge between manual tracking and real-time clinical intervention, empowering doctors with automated progress analysis.
- Alert System: Nabd tracks every physiological and vital sign with an alert system that detects if there is any potential risk that may occur.

The project can be transformed into a Software-as-a-Service (SaaS) product for hospitals and private clinics. By utilizing the repository pattern and microservices architecture, it is designed for scalability and can be integrated into existing digital health infrastructures to deepen the digitalization of the healthcare industry.

Rigorous mathematical frameworks govern the analytical core of Nabd:

- Logistic Regression: Used to model the probability of GDM based on clinical features.
- Random Forest Ensemble: Utilized for multi-class health status classification (Low, Moderate, High) through majority voting and GINI impurity minimization.
- OCR Framework: Employs the docTR deep learning framework for text detection and recognition.

References: Relying on standards from ACOG, WHO [7]-[30], and clinical datasets like Electric Sheep Africa [31], and the Figshare gestational diabetes dataset [32].

For the project component, we used multiple tools in different stacks:

- Software Stack: .NET (ASP.NET Core MVC), Python 3.10, FastAPI, Entity Framework Core, SQL Server, Bootstrap, HTML, CSS, and JavaScript.
- AI Libraries: Scikit-Learn, Pandas, docTR, and Regax for parameter extraction.
- Infrastructure: SingalIR for real-time notifications and Hugging Face for model hosting.
- Hardware: TPU v4 accelerators on Google Colab were used for model training.

Nabd focuses on 4 main stakeholders or parties for our system:

- Pregnant Women: For health tracking and product safety verification.
- Doctors: For clinical decision support and patient monitoring.
- Clinical Assistants: For streamlined scheduling and registration.
- Healthcare Facilities: For managing appointments, clinic operations, and digital health records.

Nabd demonstrates strong potential for commercialization as a health-technology startup. The combination of a medical dashboard, an AI diagnostic tool, an alert system, direct connections to clinical, an easier appointment process, and a consumer safety scanner this combination provides a differentiated value proposition. that addresses both clinical and consumer needs in the maternal health market.

The current cost of Nabd relies on low cost, focusing on development using .NET and Python 3.10, with high-performance AI training executed on TPU v4 accelerators. Nabd has a business model on a subscription-based model, where:

1) Support for medical centers and independent doctors: with a fixed subscription for both categories, but with the possibility of increasing in medical centers as the number of doctors increases.
2) AI Chatbot Integration: An advanced subscription tier that provides patients with interactive AI-driven support.

Future work will focus on extending system functionality and improving performance.
Some of our suggestions are:

- Mobile Integration: Developing native Android and IOS applications for better accessibility.
- Wearable Syncing: Integrating data from medical IoT devices for vital sign monitoring.
- Expansion: Extending the OCR capabilities to support a variety of medical documents

Team Structure and Responsibilities

Name	Contribution (%) and Description
Eman Yasser Mohamed Elnaggar	Email: emanelnaggar.2019@gmail.com Number: 01025106441 Description: Serves as the Team Lead, also responsible for implementing core features, including the AI-driven product ingredient scanner, medical lab report analysis (OCR), the intelligent chatbot, and the system-wide real-time alert engine. Contribution: 20%
Mennatullah Mohammed Mohammed Taha	Email: menat1094@gmail.com Number: 01010479373 Description: Contributed to the initial data collection for the system and played a key role in developing the AI models for Gestational Diabetes Mellitus (GDM), Maternal Risk Detection, and participated in the system-wide real-time alert engine. Contribution: 20%
Mariam Mohammed Mohammed Elnemrawy	Email: mariamelnemrawy25@gmail.com Number: 01055744840 Description: Played a pivotal role in conducting the core data research for the project and in developing the AI models for Gestational Diabetes Mellitus (GDM), early maternal risk prediction, and participated in the system-wide real-time alert engine. Contribution: 20%
Karim Yousry Abdo Helal	Email: karimhelalkh110@gmail.com Number: 01100548535 Description: Played a key role as the Team Lead overseeing system design and was effective in developing substantial portions of the system architecture and both the front-end and back-end interfaces. Contribution: 20%
Kareem Elmetwly Elmetwly Shakl	Email: kareem.shakl.eg@gmail.com Number: 01146753212 Description: Designing and implementing the system database. Led the development of the appointment scheduling system, cancellation, and availability features, and collaborated on back-end and front-end development Contribution: 20%

Partners and financiers.

Currently, Nabd does not have any partners or financiers.

Abstract

Maternal mortality and pregnancy-related complications, such as Gestational Diabetes Mellitus, remain a global health challenge [1]-[2]. Existing platforms often focus on domains like appointment scheduling or passive data logging, leaving a critical gap in real-time clinical intervention and automated risk detection. Nabd addresses these gaps by providing an integrated, AI-driven ecosystem that bridges the divide between doctors, pregnant women, and the main services they need. Nabd is intended as a clinical decision support and early warning system, not as a replacement for physician judgment

Nabd utilizes a three-tier architecture—combining a .NET back-end with Python FastAPI microservices—to offer a proactive clinical decision support system. Key technical contributions include:

- An OCR Pipeline: Used to extract data from lab reports and product ingredient labels.
- AI Models: Features a model for GDM prediction and another classifier for multi-class maternal health risk detection.
- Ingredient Safety Scanner: A specialized tool that cross-references extracted product ingredients against a medical dataset to warn of harmful substances.
- Real-Time Monitoring: Integrates SignalR for instant risk alerts and a dedicated physician's dashboard for continuous supervision.

Nabd Results highlight the system's reliability:

- Maternal Risk Detection: Achieved 98% accuracy in categorizing patients into risk levels.
- GDM Prediction: Reached 93.52% accuracy with an ROC AUC of 0.947.
- System Efficacy: The integration of automated OCR and real-time alerts transforms unstructured data into actionable clinical insights.

Nabd demonstrates that merging modern software engineering with deep learning can significantly enhance maternal safety through early risk intervention and scalable healthcare management.

Abbreviations and Terms

Abbreviation	Definition
GDM	Gestational Diabetes Mellitus
OCR	Optical Character Recognition
ROC AUC	Receiver Operating Characteristic - Area Under the Curve
MVC	Model-View-Controller
ASP	Active Server Pages
ACOG	American College of Obstetricians & Gynecologists

Introduction

Pregnancy represents a critical life stage where the health of both mother and fetus is very important. Despite its importance, maternal health risks are escalating globally. Key challenges include:

- Clinical Risks: Approximately 14% of pregnant women suffer from GDM [1], which is linked to long-term cardiovascular issues.
- Consumer Safety Gaps: products contain harmful ingredients, such as retinoids [17]-[30], yet often lack clear safety information for mothers.
- Accessibility Barriers: These risks are often exacerbated by information gaps, unhealthy lifestyles, and a lack of reliable medical alternatives when direct care is unavailable.

Addressing these issues is vital for reducing maternal risks and health complications. Beyond individual tracking, there is a critical need to assist healthcare practitioners by enabling more proactive clinical decision-making through real-time alerts and automated analysis.

While digital healthcare features several specialized platforms, they often operate in silos:

- Passive Tracking: Platforms like Flo [4] provide logging without predictive intelligence.
- Administrative Focus: Services like Vezeeta [6] prioritize scheduling and patient-provider connectivity rather than clinical decision support.
- Static Databases: Scanners like Yuka [5] are often restricted to fixed barcode databases.

Nabd differs by providing a fully integrated ecosystem that bridges the gap between manual tracking and real-time clinical intervention. Unlike predecessors, Nabd:

- Acts as a Clinical Support System: It assists doctors through continuous monitoring and proactive, AI-driven alerts.
- Employs Dynamic OCR: Using the docTR framework, the system can interpret medical lab reports or ingredient labels dynamically rather than relying on a fixed product database.
- Delivers High-Accuracy Predictions: The system features an active predictive layer with a 98% accuracy rate in maternal risk detection and 93.52% accuracy for GDM prediction.

The Nabd ecosystem is built on a scalable, three-tier architecture that integrates several advanced AI components:

- Automated OCR Pipeline: Utilizes the docTR library for extracting data from lab reports and product ingredient labels.
- GDM Prediction Engine: A Logistic Regression model that estimates the probability of Gestational Diabetes with 93.52% accuracy.
- Maternal Health Risk Detection: A Random Forest classifier that categorizes health status into Low, Moderate, and High-risk levels.
- Comprehensive Monitoring Framework: A web-based platform facilitating direct communication, clinical dashboards for doctors, and real-time notifications via SignalR.

Novelty

The core innovation of Nabd lies in its transition from manual health tracking to real-time tracking and integrating AI models into a comprehensive, multi-layered healthcare ecosystem.

1) Active Predictive Intelligence vs. Passive Tracking:

   Existing platforms act as passive logging tools; Nabd introduces an active predictive layer:

   - High-Accuracy Risk Assessment: It utilizes an AI model to classify maternal health status into Low, Moderate, or High risk with 98% accuracy.
   - Early GDM Detection: The system includes another AI model that estimates the probability of Gestational Diabetes Mellitus (GDM) with 93.52% accuracy.
   - Automated Lab Interpretation: It automates the extraction of parameters from printed test reports (e.g., CBC, HbA1c, Ferritin) [7]-[16] using Regex-based logic.
   - Clinical Threshold Alerts: A server-side evaluation engine identifies critical health boundaries (e.g., blood pressure or sugar spikes) to trigger real-time alerts.

1) Flexible OCR vs. Static Databases

   Commercial scanners, which are restricted to fixed barcode databases, Nabd offers a more flexible solution: Utilizing the docTR framework, Nabd can scan any ingredient label dynamically.

2) Integrated Three-Tier System Architecture

   Nabd is built on a robust, enterprise-grade three-layer architecture using .NET (ASP.NET Core MVC):

   - Role-Specific Dashboards: The system provides specialized interfaces for Patients, Doctors, Assistants, and Admins to ensure secure data orchestration.
   - Hybrid. NET-Python Infrastructure: It utilizes FastAPI microservices to bridge the high-performance .NET back-end with Python-based AI models via RESTful APIs.
   - Real-Time Synchronization: Through ASP.NET Core SignalR, the platform pushes critical risk notifications to the doctor's dashboard instantly without requiring a page refresh.

3) Clinical Decision Support System (CDSS)

   While platforms like Vezeeta [6] focus on administrative scheduling, Nabd acts as a clinical decision support tool:

   - Proactive Supervision: It empowers doctors with comprehensive management tools, including Digital Health Records and real-time risk notifications.
   - Operational Integrity: The system includes conflict-free appointment booking enforced at the database level and secure role-based access control (RBAC).

Why This Differentiation is Valuable

This holistic approach addresses the information gap between clinical visits. By combining high-sensitivity AI screening with a complete operational system, Nabd ensures that no high-risk case is overlooked, transforming maternal care from reactive treatment to proactive prevention.

Methodology

The Nabd platform is built on a structured three-layer architecture designed for scalability and maintainability:

- Presentation Layer: Developed using Razor Views and role-specific layouts (Patient, Doctor, Assistant, Admin) to ensure clear user context.
- Business Logic Layer: Encapsulates domain rules within Controllers and specialized Service classes (e.g., AlertService).
- Data Access Layer: Utilizes the Repository Pattern to abstract operations for over 20 entities, ensuring decoupled data management. Integration Hub: A System Gateway orchestrates request handling, while a FastAPI microservice layer bridges the .NET backend with Python-based AI models via RESTful APIs.

The project employs a high-performance, hybrid technology stack:

- Backend Framework: ASP.NET Core MVC targeting .NET 8.0.
- Database: Microsoft SQL Server managed through Entity Framework Core.
- AI Microservices: Python 3.10 with FastAPI for real-time inference.
- Machine Learning Libraries: Scikit-learn for predictive models and Pandas for data manipulation.
- Engine: The docTR deep learning framework for text detection in lab reports and ingredient labels.
- Real-time Communication: ASP.NET Core SignalR for instantaneous push notifications.
- Hardware Acceleration: Google Colab with TPU v4 accelerators for model training and inference

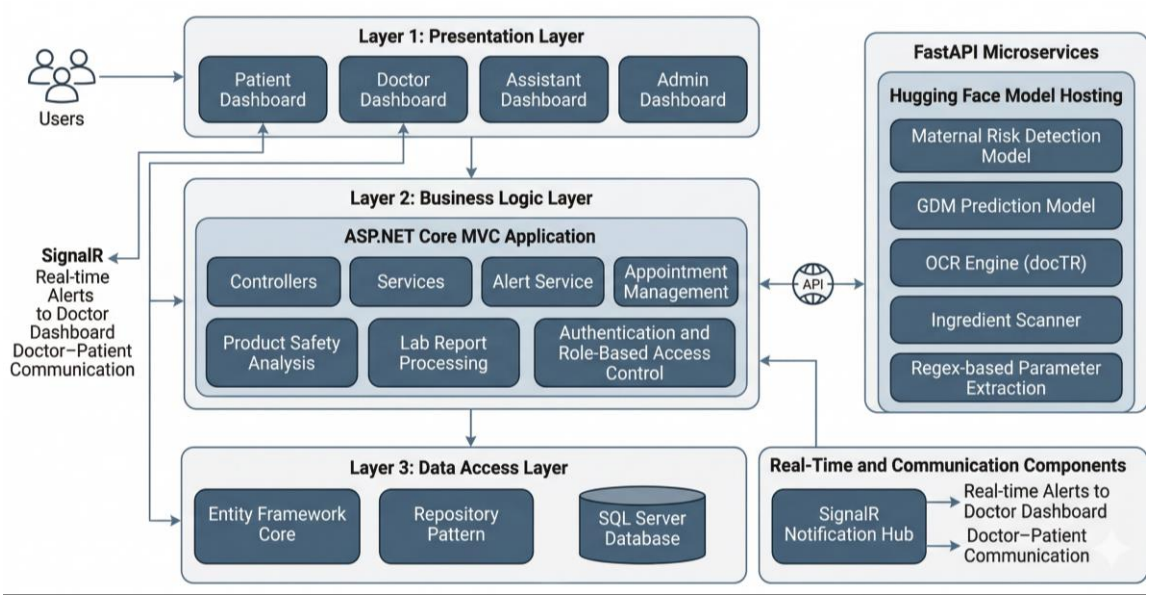

Figure 1. System Architecture of Nabd Platform

Special Cases and Logic The system incorporates specific clinical and logical handlers for complex scenarios:

- Medical Thresholding: A Rule-Based System evaluates if physiological values fall within normal ranges based on international clinical standards.
- Product Safety Mapping: Ingredients are categorized as 'Avoid' (harmful) or 'Risky' (require consultation), providing a safety guide.
- Role-Based Data Isolation: RBAC (Role-Based Access Control) ensures that users can only access routes and data relevant to their specific role, preventing cross-role exposure

To maintain high integrity, the project employs several data cleaning and validation workflows: Human-in-the-Loop Validation:

- An optional human verification step is included to confirm AI-extracted values from OCR before they are committed to the database.
- Data Standardization: Inconsistent inputs (e.g., "null", "neg", "-") are unified into a standard "negative" format to prevent processing errors.
- Imbalance Handling: Various resampling techniques were evaluated to address minority class representation in the maternal risk dataset, ensuring the model identifies high-risk cases accurately.
- Communication Reliability: AI communication channels utilize the IHttpClientFactory pattern with retry logic and exponential back-off to handle network latency or service interruptions.
- Cross-Site Request Forgery (CSRF): Protection is uniformly applied via [ValidateAntiForgeryToken] on all state-modifying endpoints

Results

The evaluation of Nabd demonstrates the clinical efficacy and technical stability of the integrated ecosystem

1) *Predictive Engine Performance*

The system's AI layer showed exceptional diagnostic reliability across both predictive engines:

- Maternal Health Risk Detection: The Random Forest model achieved an outstanding accuracy of 98%, effectively categorizing patients into High, Moderate, and Low risk levels. The high precision across all classes ensures that the Physician's Dashboard receives reliable, actionable alerts.
- GDM Prediction: The Logistic Regression engine achieved an accuracy of 93.52% and a ROC AUC of 0.947. With a Recall of 77.7%, the system prioritizes identifying potential cases for early intervention, a critical requirement in proactive screening.

2) *Automated OCR & Real-time Processing*

The docTR pipeline transformed unstructured medical data into standardized clinical records with high recognition accuracy. The integration with FastAPI enabled low-latency ingredient analysis, where the system successfully cross-referenced extracted text against the safety database to provide immediate "Avoid/Safe" classifications.

3) *System Discussion*

The results validate the—NET-Python hybrid architecture. While the GDM model’s precision (0.70) and ROC-AUC of 0.947 reflect a trade-off for higher sensitivity, this is a deliberate system design to ensure no high-risk case is overlooked. The seamless data flow—from image upload via the web interface to risk assessment and SignalR alert triggering—demonstrates the platform's readiness for real-world clinical deployment.

These results and the structure of this system greatly help us in achieving our goals of providing an integrated healthcare system between mother and patient. It also helps us in early prediction of risks in pregnancy and also of gestational diabetes, which gives an effective system based on trust and provides all services to patients, which contributes to the success of the system and attracts attention.

Product

The system is applicable in clinical and home settings for:

- Maternal Risk Detection: Identifying high, moderate, or low-risk pregnancies based on physiological parameters.
- Early GDM Screening: Predicting Gestational Diabetes Mellitus to allow for early intervention.
- Ingredient Safety: Helping expectant mothers avoid harmful chemicals in skincare and body products.
- Clinical Management: Streamlining doctor-patient communication, appointment booking, and digital health record management.

The components used in software and hardware are:

Software Components

- .NET 8.0 (ASP.NET Core MVC) for the back-end and web framework.
- Bootstrap, HTML, CSS, and JavaScript for front-end.
- Python (FastAPI) for AI microservices.
- Microsoft SQL Server for data persistence.
- docTR Library for OCR and Scikit-learn for machine learning.
- SignalR for real-time notifications.

Hardware/Electronic Components:

- TPU v4 Accelerators (Google Colab) for high-performance AI compute power during training and inference.
- Standard computing devices (PCs, laptops, or mobile devices) to access the web interface.

Currently, the project relies on low-cost approaches, focusing on development with .NET 8.0 and Python 3.10, with high-performance AI training executed on TPU v4 accelerators.

The expected cost of Nabd follows a Low-Cost/High-Efficiency model by leveraging open-source technologies and scalable cloud infrastructure:

- Development Costs: Minimized by using robust frameworks like .NET 8.0, Python (FastAPI), and SQL Server, which offer high performance without expensive licensing fees.
- Operational Costs: Primarily consist of cloud hosting (e.g., Hugging Face for model hosting, Azure or AWS for the web application) and database maintenance.
- Estimated MVP Cost: Since it is a Software-as-a-Service (SaaS) product, the initial investment is focused on server uptime and API management, making it affordable for small clinics and individual practitioners.

The Nabd system has a high potential with features, technologies, and scalability in the system to attract a big segment of the market. There is an example of our targets:

- Millions of pregnant women globally require consistent monitoring, especially in regions with limited access to immediate clinical care.
- Market Need: With approximately 14% of pregnancies [1] affected by GDM worldwide, there is a high demand for predictive tools that reduce complications.
- Competitive Edge: Unlike generic health apps, Nabd provides clinical-grade AI analysis and a direct link to healthcare providers, positioning it as a specialized medical tool rather than a lifestyle app.

The Nabd system basically focuses on three main customers:

- Patients: Pregnant women seeking continuous health tracking and product safety verification.
- Healthcare Providers: Doctors and clinics requiring a clinical decision support system with real-time risk alerts.
- Medical Assistants/Clinics: Administrative staff managing schedules and patient registration

We can represent the Nabd business model canvas with the following:

- Value Proposition: Proactive maternal care, early risk detection, and seamless doctor-patient communication.
- Customer Segments: Pregnant women, private obstetricians, and maternity hospitals/clinics.
- Revenue Streams:
  - B2B: Subscription-based model for hospitals and clinics to manage their patients.
  - B2C: Premium features for mothers, such as advanced AI health insights and a direct chatbot.
- Key Partners: Medical laboratories, pharmaceutical companies (for safe ingredient data), and cloud service providers.

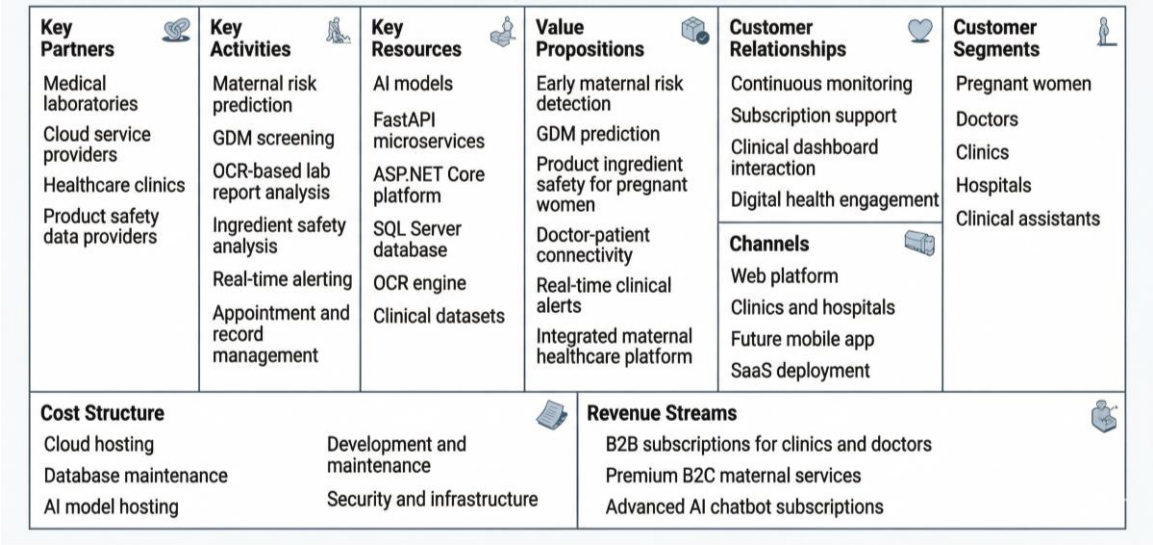

Figure 2. Business Model Canvas of Nabd

Competition is a part of any field, and what makes business success is what is the difference it offers. Nabd offers high competitive value to the main 3 competitors:

1) Vezeeta [6]: Primarily focused on clinical appointments and administrative scheduling.
2) Flo (Pregnancy Mode) [4]: A global platform used for passive maternal vital logging and symptom tracking.
3) Yuka [5]: A tool for evaluating product ingredients, though limited to a fixed barcode database compared to Nabd's dynamic OCR.

Nabd pricing strategy works as a subscription-based business model, which is based on:

1) Support for medical centers and independent doctors: with a fixed subscription for both categories, but with the possibility of increasing in medical centers as the number of doctors increases.
2) AI Chatbot Integration: An advanced subscription tier that provides patients with interactive AI-driven support.

One of the Nabd system architecture steps is to think about what risks we will face and how to solve them, and there are some risks with its solutions:

- Risk: Incorrect AI diagnosis. -Mitigation: A "human verification" step for OCR data and a specialized 0.7 probability threshold for GDM to minimize false positives.
- Risk: Data breaches. - Mitigation: ASP.NET Core Identity for secure hashing, CSRF protection, and role-based access control (RBAC).
- Risk: Clinical oversight. - Mitigation: Using a supportive, non-alarmist manner for patient alerts and ensuring doctors have a comprehensive clinical dashboard for oversight.

Nabd is built on a Three-Tier Architecture, ensuring it can grow from a local tool to a global platform:

- Technical Scaling: Developing native iOS and Android applications and integrating IoT/Wearable devices (like smartwatches) for continuous heart rate and blood pressure tracking.
- Feature Expansion: Enhancing the OCR capabilities to support a wider range of medical documents and integrating a multi-language AI chatbot for 24/7 support.
- Regional Expansion: Partnering with government health ministries to integrate Nabd into national maternal health programs, aiming to digitize and improve prenatal care at scale.

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